

MODULE 1: Laplace Transform

Definition

If $f(t)$ is a function of t satisfying certain conditions, then the definite integral

$$\phi(s) = \int_0^{\infty} e^{-st} \cdot f(t) dt$$

when it exists, is called the Laplace Transform of

$$L[f(t)] = \int_0^{\infty} e^{-st} \cdot f(t) dt$$

Example 1: Find Laplace transform of $f(t)$

where $f(t) = t$, for $0 < t < 4$

$f(t) = 5$, for $t > 4$

$$\rightarrow L[f(t)] = \int_0^{\infty} e^{-st} \cdot f(t) dt$$

$$= \int_0^4 e^{-st} \cdot f(t) dt + \int_4^{\infty} e^{-st} \cdot f(t) dt$$

$$= \int_0^4 e^{-st} \cdot t dt + \int_4^{\infty} e^{-st} \cdot 5 dt$$

$$= \left[t \left(\frac{e^{-st}}{-s} \right) - (-1) \cdot \left(\frac{e^{-st}}{s^2} \right) \right]_0^4 + 5 \left[\frac{e^{-st}}{-s} \right]_4^{\infty}$$

$$= -\frac{4}{s} e^{-4s} - \frac{1}{s^2} \cdot e^{-4s} + \frac{1}{s^2} + \frac{5}{s} e^{-4s}$$

$$= \frac{1}{s^2} + \left(\frac{1}{s} - \frac{1}{s^2} \right) \cdot e^{-4s}$$

Example 2: Find LT of $f(t)$ where $f(t) = 3$ $0 < t < 5$
 $= 0$ $t > 5$

$$\begin{aligned} \rightarrow L[f(t)] &= \int_0^{\infty} e^{-st} f(t) dt \\ &= \int_0^5 e^{-st} \cdot 3 dt + \int_5^{\infty} e^{-st} (0) dt \\ &= 3 \left[\frac{-e^{-st}}{s} \right]_0^5 \\ &= \frac{3}{s} [-e^{-5s} - e^0] \\ &= \frac{3}{s} [1 - e^{-5s}] \end{aligned}$$

Example 3: $f(t) = t$ $0 < t < 1/2$
 $= t-1$ $1/2 < t < 1$
 $= 0$ $t > 1$

$$\rightarrow L[f(t)] = \int_0^{1/2} e^{-st} \cdot t dt + \int_{1/2}^1 e^{-st} (t-1) dt + \int_1^{\infty} 0 dt$$

$$\textcircled{1} \int_0^{1/2} e^{-st} \cdot t dt$$

$$u = t \quad v = e^{-st}$$

$$t \int e^{-st} dt - \int dt \cdot t \int e^{-st} dx$$

$$t \left[\frac{e^{-st}}{-s} \right] - \int \frac{e^{-st}}{(-s)} dt$$

$$\left[t \left[\frac{e^{-st}}{-s} \right] - \frac{e^{-st}}{s^2} \right]_0^{1/2}$$

$$\int u v dx = u \int v dx - \int (du \int v dx) dx$$

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example 4 $f(t) = \begin{cases} \sin t & 0 < t < 2\pi \\ 0 & t > 2\pi \end{cases}$

$$\rightarrow L[f(t)] = \int_0^{2\pi} e^{-st} \cdot \sin t \, dt + \int_{2\pi}^{\infty} e^{-st} \cdot 0 \, dt$$

$$\Rightarrow \int_0^{2\pi} e^{-st} \cdot \sin t \, dt$$

$\int u v dx = u \int v dx - \int \frac{du}{dx} \int v dx$

$$\Rightarrow \int_0^{2\pi} \sin t \int e^{-st} dt - \int \frac{d \sin t}{dt} \int e^{-st} dt$$

$$\Rightarrow \left[\sin t \left[\frac{e^{-st}}{-s} \right] - \int \cos t \cdot \frac{e^{-st}}{-s} dt \right]_0^{2\pi}$$

$$\star \int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$\Rightarrow a \cdot \int_0^{2\pi} \sin \left(\frac{e^{-st}}{s} \right) - \int_0^{2\pi} \cos t \left(\frac{e^{-st}}{-s} \right) dt$$

$$\Rightarrow \sin t \left(\frac{e^{-st}}{s} \right) + \frac{1}{s} \int \cos t e^{-st} dt$$

$$-a = -s \quad b = 1$$

$$\left[\frac{e^{-st}}{s^2 + 1} [-s \sin x - \cos x] \right]_0^{2\pi}$$

$$\left[\frac{e^{-2\pi s}}{s^2 + 1} [-s \sin 0 - \cos 0] \right]$$

$$\Rightarrow \frac{e^{-2\pi s}}{s^2 + 1} (-1) - \frac{1}{s^2 + 1} (0 - 1)$$

Laplace Transform of some standard functions:

$$(i) L[e^{at}] = \frac{1}{s-a}$$

$$(ii) L[e^{-at}] = \frac{1}{s+a}$$

$$(iii) L[1] = \frac{1}{s}$$

$$(iv) L[k^at] = L[e^{at \log k}] = \frac{1}{s - a \log k}$$

$$(i) L[\sin at] = \frac{a}{s^2 + a^2}$$

$$(ii) L[\cos at] = \frac{s}{s^2 + a^2}$$

$$(i) L[\sinh at] = \frac{a}{s^2 - a^2}$$

$$(ii) L[\cosh at] = \frac{s}{s^2 - a^2}$$

$$(N) L[t^n] = \frac{\sqrt{n+1}}{s^{n+1}} = \frac{n!}{s^{n+1}} \text{ if } n \text{ is +ve int}$$

Properties of Laplace Transform:-

Property 1:- Linearity property

$$L[Af_1(t) + Bf_2(t)] = A L[f_1(t)] + B L[f_2(t)]$$

Example:- $L[\sin 2t + t^2 + e^{3t} - \sinh 4t]$

$$\Rightarrow L[\sin 2t] + L[t^2] + L[e^{3t}] - L[\sinh 4t]$$

$$\Rightarrow \frac{2}{s^2 + 2^2} + \frac{2!}{s^3} + \frac{1}{s-3} - \frac{4}{s^2 - 4^2}$$

Example:- $L[\sin 2t \cos 3t]$

$$\Rightarrow L[\frac{\sin 5t + \sin(-t)}{2}]$$

$$\Rightarrow \frac{1}{2} \left[\frac{5}{s^2 + 5^2} - \frac{1}{s^2 + 1^2} \right]$$

Example:- $L[\cos t \cos 2t \cos 3t]$

$$\Rightarrow \cos A \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$$

$$L\left[\frac{1}{2} (\cos 3t + \cos t) \cos 3t\right]$$

$$\frac{1}{2} L[\cos^2 3t + \cos 3t \cos t]$$

$$\frac{1}{2} \left[\frac{1 + \cos 6t}{2} \right] + \frac{1}{2} \cos 4t \cos 2t$$

Example:- $L[\sin^6 t]$

$$\Rightarrow n = \cos t + i \sin t$$

$$\frac{1}{x} = \cos t - i \sin t$$

$$n - \frac{1}{n} = 2i \sin t$$

$$\frac{1}{2i} \left(n - \frac{1}{n} \right) = \sin t$$

$$\sin^6 t = \left(\frac{1}{2i} \right)^6 \left(n - \frac{1}{n} \right)^6$$

$$\Rightarrow \left(\frac{1}{2i} \right)^6 \left[{}^6C_0 x^6 \left(\frac{1}{x} \right)^0 + {}^6C_1 n^6 \left(\frac{1}{x} \right)^1 + {}^6C_2 n^6 \left(\frac{1}{x} \right)^2 \right]$$

$$\Rightarrow \left(\frac{1}{2i} \right)^6 \left[n^6 - 6x^4 + 15x^2 - 20 + 15 \frac{1}{x^2} - \frac{6}{x^4} \right]$$

$$x^n = \cos nt + i \sin nt$$

$$\Rightarrow \left(\frac{1}{2i} \right)^6 \left[\cos 6t + i \sin 6t + \cos 6t - i \sin 6t \right]$$

$$\left(\frac{1}{2i} \right)^6 \left\{ 2 \cos 6t - 6(2 \cos 4t) + 15(2 \cos 2t) - 20 \right\}$$

$$\frac{-1}{32} \left[\frac{s}{s^2+6^2} - 6 \left(\frac{s}{s^2+4^2} \right) + 15 \left(\frac{s}{s^2+2^2} \right) - \frac{10}{s} \right]$$

Property 2:- Change of scale

If $L[f(t)] = \phi(s)$ then $L[f(at)] = \frac{1}{a} \phi\left(\frac{s}{a}\right)$

Example:- $L[f(t)] = \frac{s}{s^2 + 4s + 2}$ Find $L[f(3t)]$

Soln:- $L[f(3t)] = \frac{1}{3} \phi\left(\frac{s}{3}\right)$

$\phi(s) = \frac{s}{s^2 + 4s + 2}$

$L[f(3t)] = \frac{1}{3} \cdot \frac{(s/3)}{(s/3)^2 + 4(s/3) + 2}$

$= \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{s}{s^2 + 4s + 2}$

Example:- $L[f(t)] = \frac{1}{9} \cdot \frac{9s}{s^2 + 12s + 18}$

$= \frac{s}{s^2 + 12s + 18}$

Example:- $L[f(t)] = \frac{1}{a} \log\left(\frac{s+a}{s+b}\right)$

Soln:- $L[f(5t)] = \frac{1}{5} \phi\left(\frac{s}{5}\right)$

$\Rightarrow \frac{1}{5} \cdot \frac{1}{a} \frac{\log\left(\frac{s/5+a}{s/5+b}\right)}{(s/5+b)} \Rightarrow \frac{1}{10} \log\left(\frac{s+10}{s+15}\right)$

Property 3:- First shifting Theorem

$$L[f(t)] = \phi(s)$$

$$\text{then } L[e^{at} f(t)] = \phi(s-a)$$

$$L[e^{-at} f(t)] = \phi(s+a)$$

Example:- $L[\sin 3t] = \frac{3}{s^2 + 3^2}$

$$L[e^{2t} \sin 3t] = \frac{3}{(s-2)^2 + 3^2}$$

$$L[e^{-5t} \sin 3t] = \frac{3}{(s+5)^2 + 3^2}$$

Example:- Find $L[e^{2t} \cos^2 3t]$

Soln:- $L[\cos^2 3t] = L\left[\frac{1 + \cos 6t}{2}\right]$

$$= \frac{1}{2} L(1) + L(\cos 6t)$$

$$L[\cos^2 3t] = \frac{1}{2} \left[\frac{1}{s} + \frac{s}{s^2 + 6^2} \right]$$

$$L[e^{2t} \cos^2 3t] = \frac{1}{2} \left[\frac{1}{s-2} + \frac{(s-2)}{(s-2)^2 + 6^2} \right]$$

Example:- $L[e^{3t} \cosh 2t t^2]$

Soln:- $e^{3t} \cosh 2t t^2 = e^{3t} \left(\frac{e^{2t} + e^{-2t}}{2} \right) t^2$
 $= \frac{1}{2} (e^{5t} t^2 + e^t t^2)$

$L[e^{3t} \cosh 2t t^2] = \frac{1}{2} [L[e^{5t} t^2] + L[e^t t^2]]$

$L[t^2] = \frac{2!}{s^3}$

$L[e^{5t} t^2] = \frac{2!}{(s-5)^3}$

$L[e^t t^2] = \frac{2!}{(s-1)^3}$

$L[e^{3t} \cosh 2t t^2] = \frac{1}{2} \left[\frac{2!}{(s-5)^3} + \frac{2!}{(s-1)^3} \right]$
 $= \frac{1}{(s-5)^3} + \frac{1}{(s-1)^3}$

Example:- $L[f(t)] = \frac{s}{s^2 + 2s + 4}$

Soln:- $L[e^{3t} f(2t)]$

$L[f(2t)] = \frac{1}{2} \phi\left(\frac{s}{2}\right)$

$= \frac{1}{2} \cdot \frac{s/2}{s^2 + 2s + 4} \Rightarrow \frac{1}{4} \frac{s}{s^2 + 2s + 4}$

$\frac{1}{2} \cdot \frac{(s/2)^2 + 2(s/2) + 4}{(s/2)^2 + 2(s/2) + 4} = \frac{1}{2} \cdot \frac{s^2 + 2s + 8}{s^2 + 2s + 4}$
 $\Rightarrow \frac{s}{s^2 + 4s + 16}$

$$L[e^{3t} f(2t)] = \phi(s-a)$$

$$\Rightarrow \frac{s-3}{(s-3)^2 + 4(s-3) + 16}$$

Example:- $L\left[\frac{\sin t \cos 2t}{e^t}\right]$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$\frac{1}{2} [\sin 3t + \sin(-t)] \Rightarrow \frac{1}{2} [\sin 3t - \sin t]$$

$$L[\sin t \cos 2t] = \frac{1}{2} [L[\sin 3t] - L[\sin t]]$$

$$= \frac{1}{2} \left[\frac{3}{s^2+3^2} - \frac{1}{s^2+1} \right]$$

$$L[e^{-t} \sin t \cos 2t] \Rightarrow \frac{1}{2} \left[\frac{3}{(s+1)^2+9} - \frac{1}{(s+1)^2+1} \right]$$

Examples:- $\int_0^{\infty} e^{-st} \cos ht \cos 2t dt$

Soln $L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$

$$\cos ht = \frac{e^{ht} + e^{-ht}}{2}$$

$$L[\cos ht \cos 2t] = \frac{1}{2} (L[e^{ht} \cos 2t] + L[e^{-ht} \cos 2t])$$

$$\Rightarrow \frac{1}{2} \left[\frac{s-1}{(s-1)^2+2^2} + \frac{s+1}{(s+1)^2+2^2} \right]$$

Put $s = 1$

$$\Rightarrow \frac{1}{2} \left[\frac{1-1}{(1-1)^2+2^2} + \frac{1+1}{(1+1)^2+2^2} \right]$$

$$\Rightarrow \frac{1}{2} \left[\frac{2}{5} \right] \Rightarrow \frac{2}{10} = \frac{1}{5}$$

Example: $L[e^{-2t} \sinh 2t \sin 3t] dt$

$$L[\sinh 2t \sin 3t] = \left(\frac{e^{2t} - e^{-2t}}{2} \right) \sin 3t dt$$

$$= \frac{1}{2} (\sin 3t e^{2t} - \sin 3t e^{-2t})$$

$$= \frac{1}{2} [L(\sin 3t e^{2t}) - L(\sin 3t e^{-2t})]$$

$$= \frac{1}{2} \left[\frac{3}{(s-2)^2+9} - \frac{3}{(s+2)^2+9} \right]$$

Put $s = 2$

$$\frac{1}{2} \left[\frac{3}{(2-2)^2+9} - \frac{3}{(2+2)^2+9} \right]$$

$$\Rightarrow \frac{1}{2} \left[\frac{3}{9} - \frac{3}{25} \right]$$

$$\Rightarrow \frac{8}{75}$$

Property 4:- Effect of multiplication by t

$$L[f(t)] = \phi(s)$$

$$\text{then } L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} \phi(s)$$

Example:- $\int_0^{\infty} e^{-3t} t \sin t \, dt$

Soln:- $\int_0^{\infty} e^{-3t} t \sin t \, dt = L[f(t)]$

$$\Rightarrow \frac{1}{s} (s \sin t) = \frac{1}{s} \left(\frac{1}{s^2 + 1} \right) = \frac{1}{s(s^2 + 1)}$$

$$\frac{1}{s(s^2 + 1)} = \frac{A}{s} + \frac{B}{s^2 + 1}$$

$$\frac{1}{s(s^2 + 1)} = \frac{A}{s} + \frac{B}{s^2 + 1}$$

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Example: - Find $[L[e^{3t} \sin 2t]]]$ - 3/9/2020

Soln: - $L[\sin 2t] = \frac{2}{s^2 + 2^2} = \frac{2}{s^2 + 4}$

$L[e^{3t} \sin 2t] = \frac{2}{s - 3}$

s by $(s - 3)$ $\Rightarrow \frac{2}{(s - 3)^2 + 2^2}$

$\frac{2}{(s - 3)^2 + 2^2}$

easy way - first multiply by t

$L[t \sin 2t] = (-1)^1 \frac{d}{ds} \left(\frac{2}{s^2 + 4} \right)$

$UV - VU = 0 \Rightarrow (-1) \frac{d}{ds} \left(\frac{2}{s^2 + 4} \right) = \frac{4s}{(s^2 + 4)^2}$

$\frac{4s}{(s^2 + 4)^2}$

$(2s)^2 - (1)^2 = 4s^2 - 1$

$L[e^{3t} t \sin 2t]$ replace s by $s - 3$

$= \frac{4(s - 3)}{(s - 3)^2 + 4}$

$\frac{4(s - 3)}{(s - 3)^2 + 4}$

$\frac{4(s - 3)}{(s - 3)^2 + 4}$

Example:- $L[t e^t \cos t \cos 2t]$

$$L[\cos t \cos 2t] = \cos A \cos B = \frac{1}{2} (\cos(A+B) + \cos(A-B))$$

$$L[\cos t \cos 2t] = \frac{1}{2} L[\cos 3t + \cos t]$$

$$= \frac{1}{2} L[\cos 3t + \cos t]$$

$$= \frac{1}{2} \left[\frac{s}{s^2+3^2} + \frac{s}{s^2+1} \right]$$

$$L[t \cos t \cos 2t] = (-1) \frac{d}{ds} \left[\frac{1}{2} \left(\frac{s}{s^2+9} + \frac{s}{s^2+1} \right) \right]$$

$$\frac{s}{s^2+9} = u \cdot v \quad u = \frac{1}{s^2+9} \quad v = s \quad \frac{u}{v} = \frac{v u' - u v'}{v^2}$$

$$= \frac{s^2+9 (s)' - s (s^2+9)'}{(s^2+9)^2}$$

$$\Rightarrow \frac{s^2+9(1) - s(2s)}{(s^2+9)^2}$$

$$\Rightarrow \frac{s^2+9-2s^2}{(s^2+9)^2}$$

$$\frac{s}{s^2+1} \Rightarrow \frac{(s^2+1)(1) - s(2s)}{(s^2+1)^2} \Rightarrow \frac{s^2+1-2s^2}{(s^2+1)^2}$$

$$L[e^t t \cos t \cos 2t]$$

$$= \frac{1}{2} \left\{ \frac{s^2+9-2s^2}{(s^2+9)^2} + \frac{s^2-2s^2+1}{(s^2+1)^2} \right\}$$

Example:- $L[te^{-2t} \sinh 4t]$

Soln: $L[\sinh 4t] = \frac{1}{s^2 - 16}$

$$te^{-2t} \sinh 4t = te^{-2t} \left[\frac{e^{4t} - e^{-4t}}{2} \right]$$

$$= \frac{1}{2} [te^{2t} - te^{-6t}]$$

$$L[te^{2t}] = \frac{1}{(s-2)^2} = \frac{1}{(s-2)^2}$$

$$L[te^{-2t} \sinh 4t] = \frac{1}{2} \left[\frac{1}{(s-2)^2} - \frac{1}{(s+6)^2} \right]$$

Example:- $\int_0^{\infty} e^{-2t} + \sin^2 t \, dt$

Soln: $\int_0^{\infty} e^{-2t} + \sin^2 t \, dt$

$$L[\sin^2 t] = L\left[\frac{1 - \cos 2t}{2}\right] = \frac{1}{2} L[1 - \cos 2t]$$

$$= \frac{1}{2} \left[\frac{1}{s} - \frac{s}{s^2 + 2^2} \right]$$

$$L[+\sin^2 t] = (-1) \frac{d}{ds} \left[\frac{1}{2} \left[\frac{1}{s} - \frac{s}{s^2 + 2^2} \right] \right]$$

$$= (-1) \frac{1}{2} \left[\frac{-1}{s^2} - \left(\frac{s^2 + 2^2(1) - s(2s)}{s^2 + 2^2} \right) \right]$$

$$= \frac{-1}{2} \left[\frac{-1}{s^2} - \left(\frac{s^2 - 2s^2 + 4}{s^2 + 2^2} \right) \right]$$

$$= -\frac{1}{2} \left\{ \frac{-1}{s^2} \left[\frac{+11.4 - s^2}{(s^2 + 4)^2} \right] \right\}$$

$$L[t \sin^2 t] = -\frac{1}{2} \left\{ \frac{1}{s^2} - \frac{s^2 - 4}{(s^2 + 4)^2} \right\}$$

$$\int_0^{\infty} e^{-st} t \sin^2 t dt = \frac{1}{2} \left\{ \frac{1}{s^2} - \frac{s^2 - 4}{(s^2 + 4)^2} \right\}$$

$$\text{Put } s = 2 = \frac{1}{8} = \frac{1}{8}$$

$$\text{example:- } \int_0^{\infty} e^{-3t} t^2 \sinh 2t dt$$

$$L[\sinh 2t] =$$

$$e^{-3t} t^2 \sinh 2t = e^{-3t} t^2 \left[\frac{e^{2t} - e^{-2t}}{2} \right]$$

$$= \frac{1}{2} [t^2 e^{-t} - t^2 e^{-5t}]$$

$$L[t^2] = \frac{2}{s^3}$$

$$L[e^{-t} t^2] = \frac{2}{(s+1)^3}$$

$$L[e^{-5t} t^2] = \frac{2}{(s+5)^3}$$

$$L[e^{-3t} t^2 \sinh 2t] = \frac{1}{2} \left[\frac{2}{(s+1)^3} - \frac{2}{(s+5)^3} \right]$$

$$\mathcal{L}[e^{-3t} \sinh 2t] = \frac{1}{2} [e^{-t} - e^{-5t}]$$

$$\mathcal{L}[t^2 e^{-3t} \sinh 2t] = (-1)^2 \frac{d^2}{ds^2} \frac{1}{2} [e^{-t} - e^{-5t}]$$

$$= (1) \frac{1}{2} (-e^{-t} + e^{-5t})$$

$$\Rightarrow \frac{1}{2} (e^{-t} - e^{-5t})$$

$$\mathcal{L}[t^2 \sin 2t] = \frac{1}{2} \left[\frac{2}{(s-2)^3} - \frac{2}{(s+2)^3} \right]$$

$$\text{Put } s=2$$

$$\Rightarrow \frac{1}{(3-2)^3} - \frac{1}{(3+2)^3}$$

$$\Rightarrow \frac{1}{1} - \frac{1}{125}$$

$$\Rightarrow \frac{124}{125}$$

$$\frac{1}{125} \left[-\frac{1}{(s-2)^2} + \frac{1}{(s+2)^2} \right]$$

$$\frac{1}{125} \left[\frac{1}{(s-2)^2} - \frac{1}{(s+2)^2} \right]$$

$$\frac{1}{125} \left[\frac{1}{(s-2)^2} - \frac{1}{(s+2)^2} \right]$$

$$\frac{1}{125} \left[\frac{1}{(s-2)^2} - \frac{1}{(s+2)^2} \right]$$

Property 5:- Division by t (opp multi by t)

If $L[f(t)] = \phi(s)$

then $L\left[\frac{f(t)}{t}\right] = \int_s^\infty \phi(s) ds$

$$L\left[\frac{f(t)}{t^2}\right] = \int_s^\infty \left(\int_s^\infty \phi(s) ds\right) ds$$

Example: $L\left[\frac{\sin t}{t}\right]$

Soln:- $L\left[\frac{\sin t}{t}\right] = \int_s^\infty \sin t ds$

$$= -\cos t$$

$$L[\sin t] = \frac{1}{s^2+1}$$

$$s^2+1$$

$$L\left[\frac{\sin t}{t}\right] = \int_s^\infty \frac{1}{s^2+1} ds$$

$$= [\tan^{-1}(s)]_s^\infty$$

$$= \frac{\pi}{2} - \tan^{-1}s$$

$$= \boxed{\cot^{-1}s}$$

Example: $L\left[\frac{1-\cos t}{t}\right]$

Soln:- $L[1-\cos t] = \frac{1}{s} - \frac{s}{s^2+1}$ $s^2 = t$
 $2s ds = dt$
 $s ds = \frac{dt}{2}$

$$L\left[\frac{1-\cos t}{t}\right] = \int_0^{\infty} \frac{1}{s} ds - \int_0^{\infty} \frac{s}{s^2+1} ds$$

$$= [\log s]_s^{\infty} - \frac{1}{2} \int_s^{\infty} \frac{dt}{t^2+1}$$

$$= [\log s]_s^{\infty} - \frac{1}{2} \tan^{-1}(t)$$

$$= [\log s]_s^{\infty} - \left[\frac{1}{2} \tan^{-1}(s^2) \right]_s^{\infty}$$

$$= [\log s]_s^{\infty} - \left[\frac{1}{2} \log(s^2+1) \right]_0^{\infty}$$

$$= \frac{1}{2} [\log s^2 - \log(s^2+1)]_s^{\infty}$$

$$= \frac{1}{2} \left[\log \left(\frac{s^2}{s^2+1} \right) \right]_s^{\infty}$$

$$= \frac{1}{2} \left[\log \left(\frac{1}{1+1/s^2} \right) \right]_s^{\infty}$$

$$= \frac{1}{2} \left[\log(1) - \log \left(\frac{1}{1+1/s^2} \right) \right]$$

$$= \frac{1}{2} [0 - \log \left(\frac{s^2}{s^2+1} \right)] \Rightarrow \frac{1}{2} \log \left(\frac{s^2+1}{s^2} \right)$$

Example:- $L\left[\frac{e^{3t} - e^{2t}}{t}\right]$

Soln:- $L\left[\frac{e^{3t} - e^{2t}}{t}\right] = \lim_{s \rightarrow 0} \left[\frac{1}{s-3} - \frac{1}{s-2} \right] \frac{1}{s}$

$$= \lim_{s \rightarrow 0} \int_0^\infty \left(\frac{1}{s-3} - \frac{1}{s-2} \right) ds$$

$$= \left[\log(s-3) - \log(s-2) \right]_0^\infty$$

$$= \log\left(\frac{s-3}{s-2}\right) \Big|_0^\infty$$

$$= \log\left(\frac{s-3}{s-2}\right) \Big|_0^\infty$$

$$= \lim_{s \rightarrow 0} \left[\log(s-3) - \log(s-2) \right]$$

$$= \lim_{s \rightarrow 0} \left[\log\left(\frac{s-3}{s-2}\right) \right]$$

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$$= \lim_{s \rightarrow 0} \left[\log\left(\frac{s-3}{s-2}\right) \right]$$

Example:- $L\left[\frac{\cosh 3t \sin 2t}{t}\right] = ?$

$L[\sin 2t] = ?$

Soln: $L[\sin 2t] = \frac{2}{s^2 + 4}$

$L\left[\frac{\sin 2t}{t}\right] = \int_s^\infty \frac{2}{s^2 + 4} ds$

$= 2 \left[\frac{1}{2} \tan^{-1}\left(\frac{s}{2}\right) \right]_s^\infty$

$= \tan^{-1}(\infty) - \tan^{-1}\left(\frac{s}{2}\right)$

$= \frac{\pi}{2} - \tan^{-1}\left(\frac{s}{2}\right) = \cot^{-1}\left(\frac{s}{2}\right)$

$L\left[\frac{\cosh 3t \sin 2t}{t}\right] = L\left[\frac{e^{3t} + e^{-3t}}{2} \frac{\sin 2t}{t}\right]$

$= \frac{1}{2} \left\{ L\left(\frac{e^{3t} \sin 2t}{t}\right) + L\left(\frac{e^{-3t} \sin 2t}{t}\right) \right\}$

$= \frac{1}{2} \left\{ \cot^{-1}\left(\frac{s-3}{2}\right) + \cot^{-1}\left(\frac{s+3}{2}\right) \right\}$

Example:- $\int_0^{\infty} \left(\frac{\sin 2t + \sin 3t}{te^t} \right) dt = \frac{3\pi}{4}$

Soln:- $L\left[\frac{\sin 2t + \sin 3t}{te^t}\right]$

$\Rightarrow L\left[e^{-t} \left(\frac{\sin 2t + \sin 3t}{t} \right)\right]$

$L[\sin 2t + \sin 3t] = \frac{2}{s^2+4} + \frac{3}{s^2+9}$

$L\left[\frac{\sin 2t + \sin 3t}{t}\right] = 2 \int_s^{\infty} \frac{1}{s^2+4} ds + 3 \int_s^{\infty} \frac{1}{s^2+9} ds$

$= 2 \left[\frac{1}{2} \tan^{-1}\left(\frac{s}{2}\right) \right]_s^{\infty} + 3 \left[\frac{1}{3} \tan^{-1}\left(\frac{s}{3}\right) \right]_s^{\infty}$

$= \tan^{-1}(\infty) - \tan^{-1}(s/2) + \tan^{-1}(\infty) - \tan^{-1}(s/3)$

$= \frac{\pi}{2} + \frac{\pi}{2} - \tan^{-1}\left(\frac{s}{2}\right) - \tan^{-1}\left(\frac{s}{3}\right)$

$\Rightarrow \pi - \tan^{-1}\left(\frac{s}{2}\right) - \tan^{-1}\left(\frac{s}{3}\right)$

$L\left[\frac{\sin 2t + \sin 3t}{t}\right] = \pi - \left[\tan^{-1}\left(\frac{s}{2}\right) + \tan^{-1}\left(\frac{s}{3}\right) \right]$

$\Rightarrow \pi - \tan^{-1}\left(\frac{s/2 + s/3}{1 - s^2/6}\right)$

$\Rightarrow \pi - \tan^{-1}\left(\frac{5s}{6-s^2}\right)$

This means

$$\int_0^{\infty} e^{-st} \left(\frac{8 \sin 2t + 8 \sin 3t}{7} \right) dt = \pi - \tan^{-1} \left(\frac{5s}{6-s^2} \right)$$

Put $s=1 \Rightarrow \pi - \tan^{-1}(1) \Rightarrow \pi - \frac{\pi}{4} \Rightarrow \frac{3\pi}{4}$

Example:- $\int_0^{\infty} \frac{e^{-t} - \cos t}{t e^{4t}} dt$

Soln:- $\int_0^{\infty} \frac{e^{-4t} e^{-t} - e^{-4t} \cos t}{t} dt$

$$\int_0^{\infty} \left(\frac{e^{-5t}}{t} - \frac{e^{-4t} \cos t}{t} \right) dt$$

$$\int_0^{\infty} e^{-4t} \left(\frac{e^{-t} - \cos t}{t} \right) dt$$

$\mathcal{L}[e^{-t} - \cos t] \Rightarrow 1 - \frac{s}{s^2+1}$

$(f(t))_{\mathcal{L}} = \frac{1}{s+1} - \frac{s}{s^2+1}$ $s^2+1=t$
 $2s ds = dt$

$$\mathcal{L} \left[\frac{e^{-t} - \cos t}{t} \right] \Rightarrow \int_s^{\infty} \left(\frac{1}{s+1} - \frac{s}{s^2+1} \right) ds$$

$$\Rightarrow \int_s^{\infty} \left[\log |s+1| - \frac{1}{2} \frac{d}{ds} \log(s^2+1) \right] ds$$

$$\Rightarrow \int_s^{\infty} [\log(s+1) - \tan^{-1}(s)] ds$$

$$\Rightarrow \int_s^{\infty} \left[\log(s+1) + \log \frac{s^2+1}{2} \right] ds$$

$$\Rightarrow \log(\infty) - \log(s+1) + \frac{1}{2} \log(\infty) - \frac{1}{2} \log(s^2+1)$$

$$= \frac{1}{2} \log \left(\frac{(s+1)^2}{s^2+1} \right) \Big|_s^\infty$$

$$= \frac{1}{2} \log \left(\frac{1 + 1/s}{1 + 1/s^2} \right) \Big|_s^\infty$$

$$= \frac{1}{2} \left[\log(1) - \log \left(\frac{1 + 1/s}{1 + 1/s^2} \right) \right] \Big|_s^\infty$$

$$= \frac{1}{2} \left[0 - \log \left(\frac{(s+1)^2}{s^2+1} \right) \right] \Big|_s^\infty$$

$$= \frac{1}{2} \log \left(\frac{s^2+1}{(s+1)^2} \right) \Big|_s^\infty$$

$$\mathcal{L} \left[\frac{e^{-t} - \cos st}{t} \right] = \frac{1}{2} \log \left(\frac{s^2+1}{(s+1)^2} \right)$$

$$\int_0^\infty e^{-st} \left(\frac{e^{-t} - \cos st}{t} \right) dt = \frac{1}{2} \log \left(\frac{s^2+1}{(s+1)^2} \right)$$

Put $s=1$

$$\int_0^\infty e^{-t} \left(\frac{e^{-t} - \cos t}{t} \right) dt = \frac{1}{2} \log \left(\frac{1^2+1}{(1+1)^2} \right)$$

$$\left[\frac{(1+2) \log(1+2) - (1+2) \log(1+2)}{2} \right] \Big|_0^\infty$$

$$\left[\frac{(1+2) \log(1+2) - (1+2) \log(1+2)}{2} \right] \Big|_0^\infty$$

$$\left[\frac{(1+2) \log(1+2) - (1+2) \log(1+2)}{2} \right] \Big|_0^\infty$$

$$\left[\frac{(1+2) \log(1+2) - (1+2) \log(1+2)}{2} \right] \Big|_0^\infty$$

PROPERTY 6 :- LAPLACE OF DERIVATIVE

If $L[f(t)] = \phi(s)$

then

$$L[f'(t)] = -f(0) + s L[f(t)]$$

$$L[f''(t)] = -f'(0) - s f(0) + s^2 L[f(t)]$$

$$L[f'''(t)] = -f''(0) - s f'(0) - s^2 f(0) + s^3 L[f(t)]$$

Example: If $f(t) = \frac{\sin t}{t}$ Find $L[f'(t)]$

Soln: $L[f'(t)] = -f(0) + s L[f(t)]$

$$f(t) = \frac{\sin t}{t}$$

$$L[f(t)] = L\left[\frac{\sin t}{t}\right] = \cot^{-1}(s)$$

$$L[f'(t)] = -f(0) + s \cot^{-1}(s)$$

$$f(0) = -1 + s \cot^{-1}(s)$$

Example: $f(t) = \begin{cases} 5 & 0 \leq t \leq 3 \\ 0 & t > 3 \end{cases}$ Find $L[f(t)]$ & $L[f'(t)]$

Soln: $L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$

$$= \int_0^3 e^{-st} (5) dt = (5) \left(\frac{e^{-3s}}{-s} \right) - \left(\frac{e^{-0s}}{-s} \right)$$

$$= \frac{5}{s} (e^{-3s} - 1) = \frac{5}{s} (1 - e^{-3s})$$

$$L[f'(t)] = -s + s \cdot 5/s (1 - e^{-3s})$$

$$= -s + 5 - s e^{-3s} \Rightarrow -s e^{-3s}$$